

## Top-K Youtube Videos

Functional requirements;

User should be able to open top-K videos based on his preference,

top-K videos should get adjusted after every 5 seconds

top-K videos should get populated based on the user's activity

Non-functional requirements:

data-partitioning,

load balancing,

fault tolerance,

vertical scalability,

horizontal scalability,

low latency,

high throughput,

CAP theorem (Here eventual consistency is fine, but here we need 99.99% uptime

i.e Availability >>> Consistency)

Core Entities:

user,

videos,

history,

activity

API :

POST:/upload

{

    userID,

    filemetadata

}

GET:/search/top/{K}

HLD:

